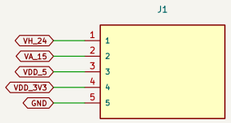
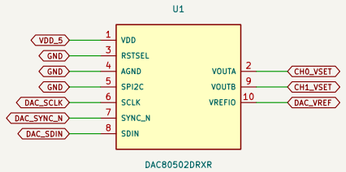
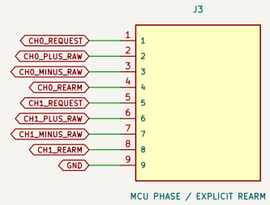
ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.







Source0_analog

File: source0.kicad_sch

Source0_permission

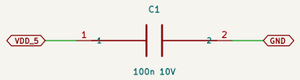
File: permission0.kicad_sch

Source1_analog

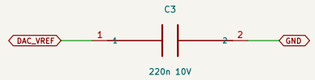
File: source1.kicad_sch

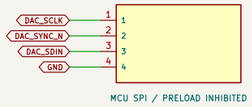
Source1_permission

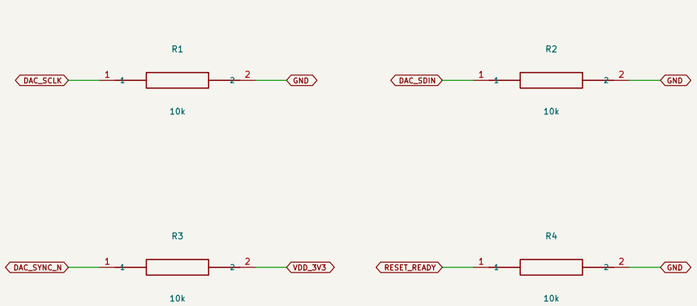
File: permission1.kicad_sch

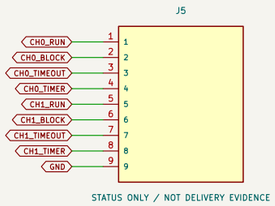


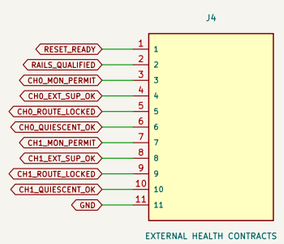












DAC initialization before READY: GAIN[0x04]=0x0103, reference divider 2 and both output buffer gains 2 => nominal 2.5 V FSR.

Code 7864 -> 2.999878 mA nominal through 100 ohm. Zero code is not terminal disconnection. Read README and external contracts.

RESET_READY held LOW through startup; REARM is an explicit pulse with REQUEST/TIMER LOW and independently observed quiescence.











Native circuit slice; artificial loads only; no PCB or implant qualification

Project ATO

Sheet: /

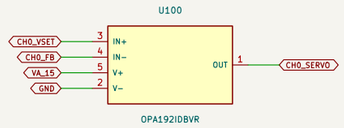
File: current_driver.kicad_sch

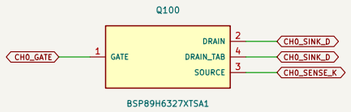
Title: ATO S4 / Dual setpoint and external contracts

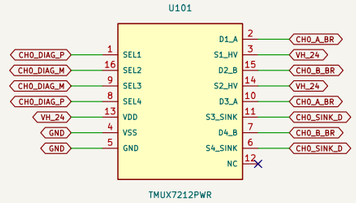
Size: A3	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/5

ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

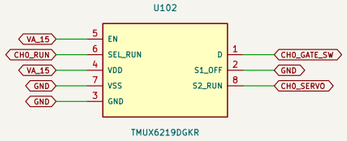
External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.

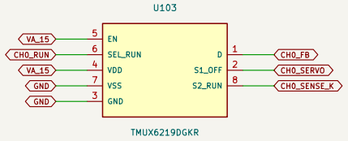


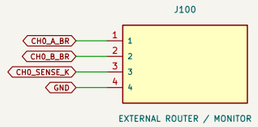




Current path: VH -> upper switch -> artificial pair load + external 2 ohm monitor -> lower switch -> BSP89 -> 100 ohm -> GND.

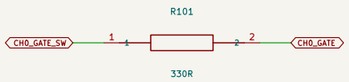


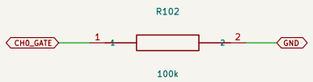


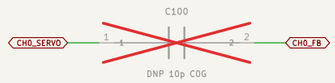


RUN low grounds gate and closes local unity feedback. RUN high selects current loop. DNP compensation needs a transient study.

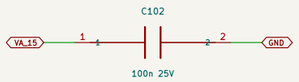


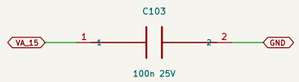






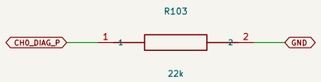


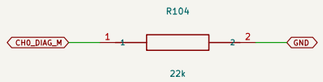














Bridge uses captured direction; all four commanded off during inhibit. Analog rail clamps, input bias and stuck switches remain real paths.

Native circuit slice; artificial loads only; no PCB or implant qualification

Project ATO

Sheet: /Source0_analog/
File: source0.kicad_sch

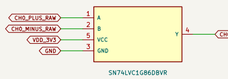
Title: ATO S4 / Source 0: regulated sink and polarity bridge

Size: A3	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 2/5

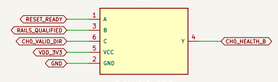
ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.







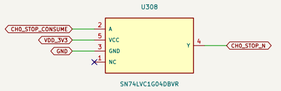




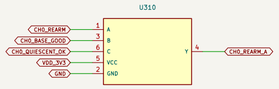








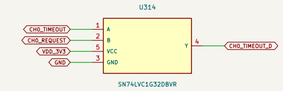




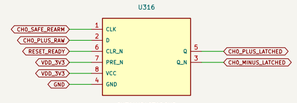




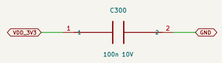


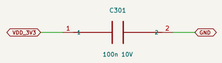












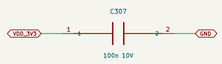


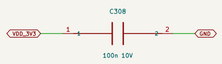












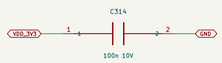


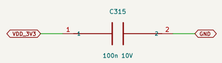












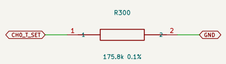
































BASE_GOOD=MON & EXT & ROUTE & READY & RAILS & XOR(raw directions). Raw REQUEST clocks TIMER: SAFE_REARM preloads direction while off.

Normal request end consumes BLOCK; TIMEOUT records held request at timer fall. REARM requires request/timer LOW + QUIESCENT_OK.

Static/event checks do not qualify asynchronous hazards, supply ramps, propagation delay or real current tails. No automatic retry.

Native circuit slice; artificial loads only; no PCB or implant qualification

Project ATO

Sheet: /Source0_permission/

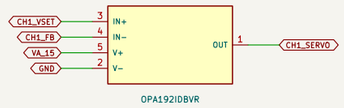
File: permission0.kicad_sch

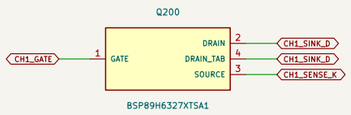
Title: ATO S4 / Source 0: independent duration and sticky admission

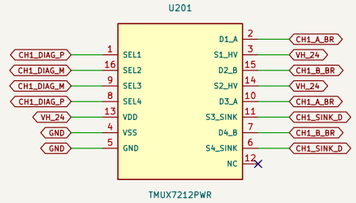
Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 3/5

ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

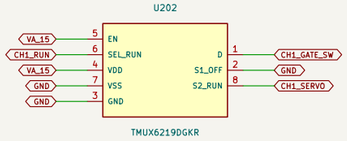
External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.

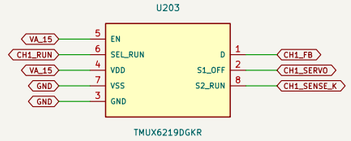


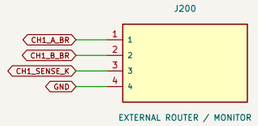




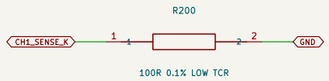
Current path: VH -> upper switch -> artificial pair load + external 2 ohm monitor -> lower switch -> BSP89 -> 100 ohm -> GND.



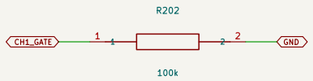


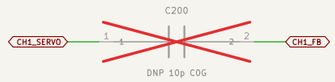


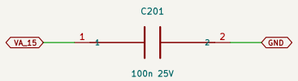
RUN low grounds gate and closes local unity feedback. RUN high selects current loop. DNP compensation needs a transient study.

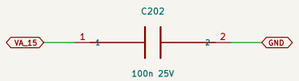


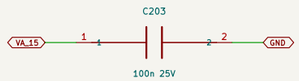








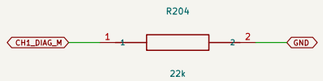














Bridge uses captured direction; all four commanded off during inhibit. Analog rail clamps, input bias and stuck switches remain real paths.

Native circuit slice; artificial loads only; no PCB or implant qualification

Project ATO

Sheet: /Source1_analog/
File: source1.kicad_sch

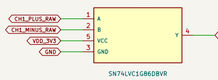
Title: ATO S4 / Source 1: regulated sink and polarity bridge

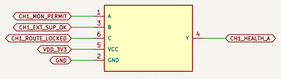
Size: A3	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 4/5

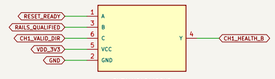
ARTIFICIAL LOADS ONLY / RESTRICTED 24 V, 3 mA, 200 us DRIVER SLICE

External rails, recording AFE, 16-contact routing, compliance, charge guard, watchdog, PCB and implant qualification remain outside.



















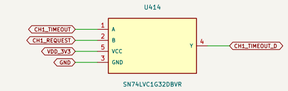


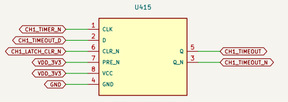


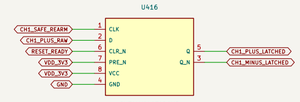




















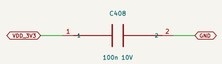














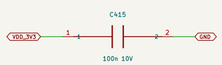


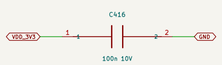










































BASE_GOOD=MON & EXT & ROUTE & READY & RAILS & XOR(raw directions). Raw REQUEST clocks TIMER: SAFE_REARM preloads direction while off.

Normal request end consumes BLOCK: TIMEOUT records held request at timer fall. REARM requires request/timer LOW + QUIESCENT_OK.

Static/event checks do not qualify asynchronous hazards, supply ramps, propagation delay or real current tails. No automatic retry.

Native circuit slice; artificial loads only; no PCB or implant qualification

Project ATO

Sheet: /Source1_permission/

File: permission1.kicad_sch

Title: ATO S4 / Source 1: independent duration and sticky admission

Size: A2	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 5/5